# Manindra de Mel

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## Education

**Australian National University** **Jan 2022 – Jun 2026**

Bachelor of Advanced Computing | Major in Machine Learning, Minor in Mathematics

Coursework: functional programming, logic & proofs, array programming, systems, machine learning. ANU Extension — Astrophysics (2021).

## Work Experience

**Rocket Remit (Frontend Developer / Data Analyst)** **Aug 2023 – present**

- Built customer-facing features in Flutter (mobile) and Angular (web) for the remittance product covering 30+ corridors across Asia, Africa, and the Pacific.
- Maintained the Groovy / Java backend and supporting SQL databases; owned DevOps for build, release, and deployment pipelines.
- Designed an ML transaction-monitoring system flagging suspicious activity across remittance flows — integrated into compliance review and reduced manual triage volume by ~40%.
- Analysed 100k+ historical transactions to surface corridor-specific risk patterns; outputs used to tune compliance rules.

**Purse (Lead Engineer)** **2025 – present**

- Building an Australian finance app aggregating super, banking, and investments from 100+ ADIs via CDR / Open Banking into a single net-worth view.
- Mobile-first Flutter client; transactions auto-categorise on-device across 20+ categories with daily-spend feeds and budget envelopes.

**Stomble (Backend Engineer)** **Nov 2022 – May 2023**

- Led a backend team of 4; designed services connecting native client apps to AWS-hosted infrastructure across 10+ microservices.
- Owned AWS configuration, deployment, and operations; raised unit-test coverage to 80%+ and onboarded 5+ engineers.

**Canberra Websites (Founder / Engineer)** **2021 – present**

- Full-stack web, data pipelines, scrapers, and AI integrations for Canberra Cloud Kitchen, Our Tutor, Quintet Automotive, Harin's Teas, and mHits AU.
- Co-built an AI-driven surgical scheduling tool (team101.life).

## Select Projects

**PlantCLEF 2026 — Co-author, accepted to CLEF 2026 [LINK]**

- ANU submission to PlantCLEF 2026 (Raj, de Mel, Wasif, Brake); multi-label species ID across 7,806 long-tailed classes at vegetation-quadrat scale.
- Partial fine-tuning of BioCLIP 2.5 ViT-H/14: only the last 4 transformer blocks unfrozen (lower 28 frozen to preserve the Tree-of-Life prior) with auxiliary cross-entropy heads at genus and family levels.
- Trained on a 2.65M-image enlarged manifest (PlantCLEF 2024 + research-grade iNaturalist pull); inference uses a 4×4 tile grid at 224 and 336 px, class-prior logit adjustment, and an adaptive probability threshold.
- Best public Macro F1 of **0.41826** (private 0.40283); paper appears in CLEF 2026 proceedings, Jena, September 2026.

**QEM Mesh Simplification with Skeletonization & CNNs [LINK]**

- Structure-aware mesh decimation preserving fine geometric features (2024).

**Real-Time Engine Tuning via ML [LINK]**

- Live optimisation of internal-combustion engine parameters for efficiency and emissions (2024).

**OpenBMB · ChatDev [LINK]**

- Multi-agent LLM software-development framework (26k+ stars); contributed deployment and CI tooling to the open release (2023).

**PPO Reinforcement Learning for Equities Trading**

- Resource-efficient policy-gradient agent; productised as a trading-bot service (2023).

## Achievements

- PlantCLEF 2026 — paper accepted to CLEF 2026 proceedings, Jena (co-author, 2026)
- IMC Global Trading Challenge (2025)
- Jane Street Electronic Trading Challenge (2024)
- AI Hack Melbourne — 4th place (2023)
- Australian National Python Convention (PyCon AU) — Speaker (2021)
- Young ICT Explorers — Regional Winner (2021)
- Code Quest, Lockheed Martin — Runners-up (2021)
- Australian Informatics Olympiad — Credit (2019–2021)

## Relevant Skills

- Python, Rust, C / C++, Haskell, TypeScript, APL, Assembly
- Machine learning, deep learning, computer vision (PyTorch)
- Reinforcement learning (PPO, policy gradients)
- Foundation model fine-tuning (ViTs)
- Full-stack web (React, Svelte, Vue, React Native, Firebase, Node)
- AWS, GCP, Linux, CI/CD
- Databases: SQL, NoSQL, Firestore
- Functional, procedural, logic, OO paradigms
- Web scraping, data pipelines, ETL

## Relevant Interests

Building computers; programming software, websites, and machine learning models; researching mathematical concepts related to computer science; mechanics & working on my vehicle.